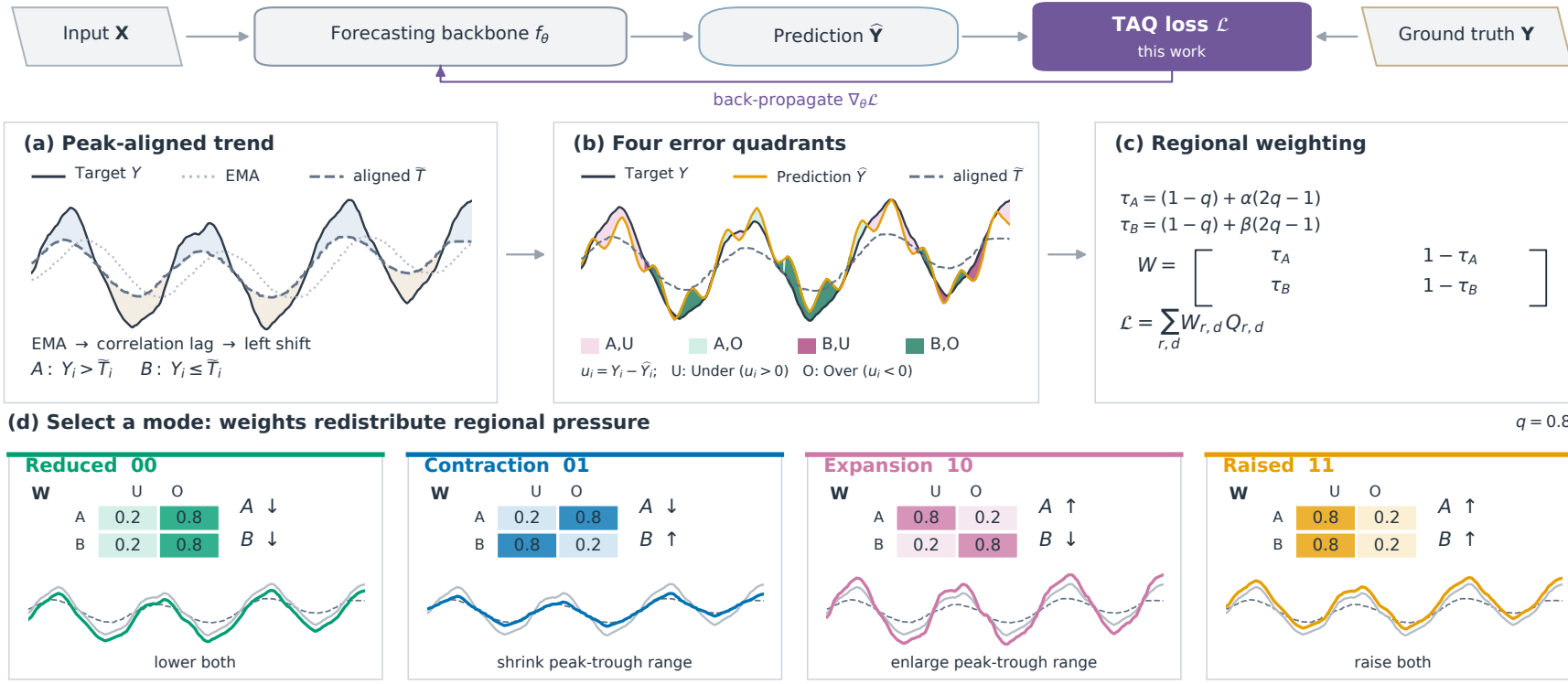

Input X
Forecasting backbone f_θ
Prediction $\hat{\mathbf{Y}}$
TAQ loss $\mathcal{L}$
this work
Ground truth Y
back-propagate $\nabla_\theta \mathcal{L}$
(a) Peak-aligned trend
Target Y
EMA
aligned $\bar{T}$
EMA → correlation lag → left shift
$A: Y_i > \bar{T}_i$ $B: Y_i \le \bar{T}_i$
(b) Four error quadrants
Target Y
Prediction $\hat{Y}$
aligned $\bar{T}$
A,U
A,O
B,U
B,O
$u_i = Y_i - \hat{Y}_i$; U: Under ($u_i > 0$) O: Over ($u_i < 0$)
(c) Regional weighting
$\tau_A = (1-q) + \alpha(2q-1)$
$\tau_B = (1-q) + \beta(2q-1)$
$W = \begin{bmatrix} \tau_A & 1-\tau_A \\ \tau_B & 1-\tau_B \end{bmatrix}$
$\mathcal{L} = \sum_{r,d} W_{r,d} Q_{r,d}$
(d) Select a mode: weights redistribute regional pressure
$q = 0.8$
Reduced 00
W U O
A 0.2 0.8 A ↓
B 0.2 0.8 B ↓
lower both
Contraction 01
W U O
A 0.2 0.8 A ↓
B 0.8 0.2 B ↑
shrink peak-trough range
Expansion 10
W U O
A 0.8 0.2 A ↑
B 0.2 0.8 B ↓
enlarge peak-trough range
Raised 11
W U O
A 0.8 0.2 A ↑
B 0.8 0.2 B ↑
raise both